

3583 PINK1 PARKINSON'S DISEASE IN NEW ZEALAND

¹Joseph Donnelly*, ¹Christina M Buchanan, ¹Shilpan G Patel, ²Garcia Vailahi, ³Satupaitea Viali, ¹Viswas Dayal, ²Richard H Roxburgh. ¹Auckland City Hospital, Auckland, New Zealand; ²University of Auckland, Auckland, New Zealand; ³Oceania University of Medicine Samoa, Apia, Samoa

10.1136/bmjno-2025-ANZAN.130

Background PINK1 Parkinson's disease has been described in Polynesian populations, however, clinical characteristics of this condition has not been well elucidated.

Methods We performed a retrospective chart review of demographic and clinical factors including age of onset of symptoms, years since onset, levodopa daily dose at time of assessment, and additional Parkinson's disease medications.

Results Amongst 27 patients with genetically confirmed PINK1 Parkinson's disease, 10 were female, the median age was 56 (range 30–70), the median age of onset was 35 (range 12–54) and the median body mass index was 31.4 (range 27 to 39). In terms of ethnicity, 15 were Samoan, 8 Tongan, 3 Tokelauan and 1 was Filipino. At the time of latest assessment, a median of 17 years (range 2 to 40) had elapsed since onset of symptoms. At this time, medication information was available on 25 of the 27 patients. 23 were on levodopa (11 monotherapy, 12 in combination with other treatments). The most common other treatments were apomorphine (n=7) and amantadine (n=5). The median levodopa daily dose (at median of 17 years since onset) was 600mg (interquartile range 350mg to 1000mg).

Conclusion The cohort of PINK1 patients in New Zealand are of almost exclusive western Polynesian ethnicity. The onset of symptoms was largely before age 50 and the patients remained on relatively low daily levodopa doses 17 years after first symptoms. This may indicate a prolonged treatment responsiveness of PINK1 Parkinson's disease.

3584 CASE REPORT. ESSENTIAL THROMBOCYTHEMIA: A RISK FACTOR FOR IIH

^{1,2}Sarah K Morgan*, ³Matthew Wright, ⁴William H Morgan, ^{1,2,5}Neha Irani. ¹Neurology, Royal Perth Hospital, Perth, WA, Australia; ²Ophthalmology department, Sir Charles Gairdner Hospital, Perth, WA, Australia; ³Haematology, Fiona Stanley Hospital, Perth, WA, Australia; ⁴Lions Eye Institute, Perth, WA, Australia; ⁵Neurology, Joondalup Health Campus, Perth, WA, Australia

10.1136/bmjno-2025-ANZAN.131

Case We present a case of an 18 year-old non-obese woman with a four-week history of headaches, and blurred vision. Although she had a remote history of acne treated with Roaccutane, the latter was ceased six months prior. Her examination was notable for a normal body mass index (21kg/m²) with Frisen grade four papilloedema.

Investigations CSF opening pressure was 40.5cmH₂O with normal constituents. MRI brain and venogram demonstrated features of raised intracranial pressure, without venous sinus thrombosis. Given her normal body habitus, secondary causes of raised ICP were explored and aside from long-standing high platelet count of 612x10⁹/L, none identified. Subsequently Jak2v617f mutation was detected.

Diagnosis The patient's clinical course was in keeping with IIH secondary to ET given the rarity of ET in this demographic, absence of typical risk factors for IIH and response to ET specific treatment.

Progress Initially, her ICP remained high despite high dose acetazolamide and serial lumbar punctures. Upon treatment with hydroxyurea, aspirin and peginterferon, there was reduction in the platelet count with simultaneous improvement in papilloedema. Acetazolamide was ceased. At one year, she remained asymptomatic on ongoing treatment for ET.

Conclusion/Discussion IIH in the context of ET without venous sinus thrombosis has been reported. Previously postulated mechanisms include partially occlusive venous sinus or impaired CSF absorption at arachnoid granulations. Herein, we hypothesize that congestion of the glymphatic system especially in the presence of a myeloproliferative disorder possibly resulted in impaired drainage of the dural venous sinuses triggering the hydrodynamic cascade of IIH.

3585 MIGRAINE AFTER MILD TRAUMATIC BRAIN INJURY: CURRENT TREATMENT APPROACHES AND RESEARCH CHALLENGES

^{1,2}Rowena Mobbs*, ³Susan Tomlinson, ¹Fatima Nasrallah, ^{4,5}Peter Goadsby, ⁶Virginia Newcombe, ⁷Bronwyn Jenkins, ^{8,9}Clare Fraser, ¹⁰Jenna Ziebell, ^{11,12}Robert Adam, ¹³Alessandro Fois, ¹⁴Anthony Mobbs. ¹Queensland Brain Institute, North Sydney, NSW, Australia; ²Mater Hospital Sydney, Sydney, NSW, Australia; ³Department of Neurology, St Vincent's Hospital, Sydney, NSW, Australia; ⁴Neurology, King's College – NIHR-Wellcome Trust King's Clinical Research Facility, London, UK; ⁵University of California, Los Angeles, U.S.A; ⁶Emergency and Urgent Care Research in Cambridge (EURECA), PACE Section, Department of Medicine, Cambridge University, Cambridge, UK; ⁷Department of Neurology, Royal North Shore Hospital, Sydney, NSW, Australia; ⁸Faculty of Health and Medicine, Save Sight Institute, University of Sydney, Sydney, NSW, Australia; ⁹Macquarie Ophthalmology, School of Clinical Medicine, Macquarie University, NSW, Australia; ¹⁰Wicking Dementia Research and Education Centre, University of Tasmania, Hobart, TAS, Australia; ¹¹Centre for Health Services Research, The University of Queensland, Brisbane, QLD, Australia; ¹²Centre for Clinical Research, The University of Queensland, Brisbane, QLD, Australia; ¹³Department of Neurology, Macquarie University Hospital, Macquarie University, NSW, Australia; ¹⁴Department of Psychology, Macquarie University, NSW, Australia

10.1136/bmjno-2025-ANZAN.132

Traumatic brain injury global prevalence is estimated to be over 55 million, with a direct economic primary care cost in Australia for mild traumatic brain injury (mTBI) of around \$50 million annually, and headache and migraine after mild traumatic brain injury (mTBI) are commonly seen in around 40% of cases. Divided into subcategories of Acute Post-Traumatic Headache and Persistent Post-Traumatic Headache (APTH/PPTH) under the International Classification of Headache Disorders (ICHD-3), significant gaps remain in our understanding of APTH/PPTH pathophysiology and its management. The molecular mechanisms of mTBI (from both direct and indirect injury) as well as clinical outcomes are wide-ranging, and as yet there are no clinical biomarkers to differentiate cerebral related structural inflammation or injury from APTH/PPTH related neurological dysfunction. Further, there are no direct neural treatments available for post mTBI, and current approaches remain largely supportive under a coordinated multidisciplinary framework, including vestibular therapy, symptom-limited physical therapy, and cognitive behavioural therapy. By contrast, pharmaceutical options are readily available for episodic and chronic migraine include amitriptyline, pizotifen, topiramate, propranolol and candesartan, therapies targeting CGRP (monoclonal antibodies and receptor antagonists) onabotulinum toxin A, or greater occipital perineural steroid injection, and these may be underutilised after mTBI. The Guidelines of the International Headache Society for controlled trials of pharmacological preventive treatment for